thomasamuti-cpu / cars_accidents_project

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

Watch 0

This github repository is a joint submission for the Moringa Phase 4 project on predicting the primary contributory cause of a car accident, given information about the car, the people in the car, the road conditions and other factors.

0 stars 0 forks 0 watching Branches Activity
Tags

Public repository

m... 4 Branches 0 Tags Go to file Go to file Add file Code

munene-gitonga Final commit 0b603f6 · 5 minutes ago

Figures	Data Modelling, Evaluation and Tuning, Re...	1 hour ago
.DS_Store	README.md, Tableau Dashboard	27 minutes ago
.gitignore	Data Modelling, Evaluation and Tuning, Re...	1 hour ago
Beyond the Crash - powerpoint.pptx	Beyond the Crash ppt , notebook.pdf, pow...	1 hour ago
Dashboard on the Primary Contribu...	README.md, Tableau Dashboard	27 minutes ago
README.md	Updated README.md	10 minutes ago
car_crash_project.ipynb	Data Modelling, Evaluation and Tuning, Re...	1 hour ago
notebook.pdf	Beyond the Crash ppt , notebook.pdf, pow...	1 hour ago
powerpoint.pdf	Beyond the Crash ppt , notebook.pdf, pow...	1 hour ago

README

1.0 Project Overview

1.1 Real Life Problem

Traffic accidents are a common and persistent public safety issue worldwide, occurring daily in both urban and rural settings. According to the World Health Organization, road traffic accidents result in over a million fatalities each year, with many people experiencing both health and financial losses. In major cities, these accidents can be contributed by either:

- Human related factors such as distracted driving, speeding or impaired driving.
- Vehicle related factors such as vehicle defect and vehicle model.
- Environmental related factors such as poor weather, low visibility and road surface issues such as poorly maintained roads.

Understanding these factors is essential to reducing traffic accidents and developing effective safety measures.

1.2 Invested stakeholders

Several groups might benefit significantly from the traffic accidents data and insights drawn from understanding the primary contributory causes. These might include:

1. Government and city authorities - who might use this to identify high risk areas and improve on road infrastructure while implementing policies aimed at reducing accidents.
2. Public safety organizations - such as the Vehicle Safety Board who might use the data to aid in designing regulation and safety campaigns and enforcement strategies based on observed trends.
3. General public and drivers - who might benefit from better awareness of risk factors.

4. Vehicle manufacturers - who might leverage on insights to improve car safety features such as weather detection systems.

2.0 Business Understanding

2.1 Business Problem

Large cities such as the City of Chicago experience frequent traffic accidents often. To address this challenge, there is need for a predictive model that can identify **primary contributory cause of car accidents** using key influencing factors including:

1. Human Factors such as sex and age.
2. Vehicle Factors such as vehicle make.
3. Environmental Factors such as weather patterns.
4. Temporal Factors such as Date and Time of the Accidents.
5. Road Design Characteristics such as road defects and road surface conditions.

Such a model would support data driven decision making aimed at reducing accident rates and improving overall road safety.

2.2 Business Questions

The project seeks to answer the following business questions:

1. What are the most common primary contributory causes of traffic accidents?
2. How do human characteristics contribute to accident occurrence?
3. How do vehicle characteristics contribute to accident occurrence?
4. How do environmental conditions such as weather influence accident occurrence?
5. Are there specific times when accidents are more likely to occur?
6. How does road design impact the likelihood of accidents?
7. Can we accurately predict the primary cause of an accident given the human, vehicle, environmental, temporal and road design characteristics?

2.3 Business Objectives

The main objectives of this project are to:

1. Develop a predictive model that can classify the primary contributory cause of traffic accidents.
2. Identify key factors (Human, Vehicle, Environmental, Temporal) that significantly influence accident occurrence.

3.0 Data Understanding

3.1 Data Description and Sources

- This project utilized 3 datasets that contain information on the accident occurence, characteristics of people involved as well as the vehicle characteristics. These are:

1. City of Chicago Accident Dataset - sourced from https://data.cityofchicago.org/Transportation/Traffic-Crashes-Crashes/85ca-t3if. This dataset has 1.04 million rows and 48 columns. Each row is a traffic crash with the row identifier as the CRASH_RECORD_ID.
2. Vehicle Accident Dataset sourced from https://data.cityofchicago.org/Transportation/Traffic-Crashes-Vehicles/68nd-jvt3. It has 2.28 million rows and 29 columns. Each row is a person with the row identifier as the PERSON_ID. The dataset also has CRASH_RECORD_ID.
3. Driver/Passenger Accident Dataset sourced from https://data.cityofchicago.org/Transportation/Traffic-Crashes-People/u6pd-qa9d. It has 2.12 million rows with 71 columns. Each row is a vehicle with the row identifier as the CRASH_UNIT_ID. This dataset however also has a CRASH_RECORD_ID.

Note: Our goal is to merge the three datasets using the common CRASH_RECORD_ID, creating a unified dataset that combines all relevant features for our analysis.

From the merger, we aim to use the following columns from the following datasets:

1. City of Chicago Dataset:

- Weather condition - Weather condition at time of crash.
- Lighting condition - Light condition at time of crash.
- Road defects - Road defects, as determined by reporting officer.
- Crash type - A general severity classification for the crash. Can be either Injury and/or Tow Due to Crash or No Injury / Drive Away.
- Primary Contributory Cause - The factor which was most significant in causing the crash.
- Crash hour - The hour of the day component of CRASH_DATE.
- Crash day - The day of the week component of CRASH_DATE.
- Crash month - The month component of CRASH_DATE.

2. Vehicle Dataset:

- Number of passengers - Number of passengers in the vehicle. The driver is not included.
- Make - The make (brand) of the vehicle.
- Vehicle Use - The normal use of the vehicle.
- Vehicle Defect - Defect the vehicle had.
- Exceeds speed limit - Whether vehicle exceeded speed limit.

3. Passenger Dataset:

- Person type - Type of roadway user involved in crash.
- Sex - Gender of person involved in crash.
- Age - Age of person involved in crash.
- Driver Vision - What, if any, objects obscured the driver's vision at time of crash.
- Physical Condition - Driver's apparent physical condition at time of crash.
- Driver Action - Driver action that contributed to the crash.
- Pedpedal Action - Action of pedestrian or cyclist at the time of crash.

3.2 Target Variable

- Our target variable is: Primary contributory cause which is a multiclass classification problem. There are 40 values of the aforementioned target variable and we aim to predict the most common primary contributory causes based on our feature variables.

3.3 Feature Variables

- All the other aforementioned variables other than primary contributory cause will act as our feature variables. They will be specifically selected following a merger of the 3 Datasets.

3.4 Challenge with large samples of rows

Following the merger, we expect 2 challenges:

1. Since a single crash may involve more than just the driver, each CRASH_RECORD_ID could correspond to multiple people. To simplify the analysis, we will focus on drivers, examining each crash event from the perspective of the drivers involved.
2. A single crash can involve multiple vehicles, so if a crash involves two cars with one person in each, it would result in at least two rows —one for each individual involved in the same crash.
3. The dataset contains millions of rows, which may exceed available computing resources. To manage this, we will limit our analysis to approximately 500,000 randomly selected samples after data cleaning, which will also help reduce the dataset size naturally.

4.0 Data Preparation and Cleaning

4.1 Data Preparation

We used several libraries to prepare and analyse the dataset:

1. Pandas - to read our csv files and perform exploratory data analysis.
2. Matplotlib - for visualizations.
3. Scikit-learn – for model fitting and evaluation. The models used were:

- Decision Forest Classifier – Base model to capture complex patterns in the dataset.
- Random Forest Classifier - To handle high dimensional data.
- Neural Networks – To handle situations of class imbalance.

4.2 Data Cleaning

Prior to data analysis we selected the following columns from the following datasets:

1. City of Chicago Dataset:

- Lighting condition - Light condition at time of crash.
- Weather condition - Weather condition at time of crash.
- Road defects - Road defects, as determined by reporting officer.
- Crash type - A general severity classification for the crash. Can be either Injury and/or Tow Due to Crash or No Injury / Drive Away.
- Primary Contributory Cause - The factor which was most significant in causing the crash.
- Crash hour - The hour of the day component of CRASH_DATE.
- Crash day - The day of the week component of CRASH_DATE.
- Crash month - The month component of CRASH_DATE.

2. Vehicle Dataset:

- Number of passengers - Number of passengers in the vehicle. The driver is not included.
- Make - The make (brand) of the vehicle.
- Vehicle Use - The normal use of the vehicle.
- Vehicle Defect - Defect the vehicle had.
- Exceeds speed limit - Whether vehicle exceeded speed limit.

3. Passenger Dataset:

- Person type - Type of roadway user involved in crash.
- Sex - Gender of person involved in crash.
- Age - Age of person involved in crash.
- Driver Vision - What, if any, objects obscured the driver's vision at time of crash.
- Physical Condition - Driver's apparent physical condition at time of crash.
- Driver Action - Driver action that contributed to the crash.
- Pedpedal Action - Action of pedestrian or cyclist at the time of crash.

Final Output:

- All 3 data sets were merged with a total of 21 columns and 1,192,524 entries.
- Dataset were filtered to contain ONLY the data that focused on drivers.
- 'Exceeding speed limit' and 'pedestrian action' columns were dropped since they contained > 95% missing data.
- The column 'Age' was also noted to have a large number of missing values which was 26.6% of total data.
- The column also had negative values.
- Filling the missing values would have led to erroneous results.
- Therefore both the missing and negative values were dropped.

The final total entries were: 858,550 rows and 21 columns.

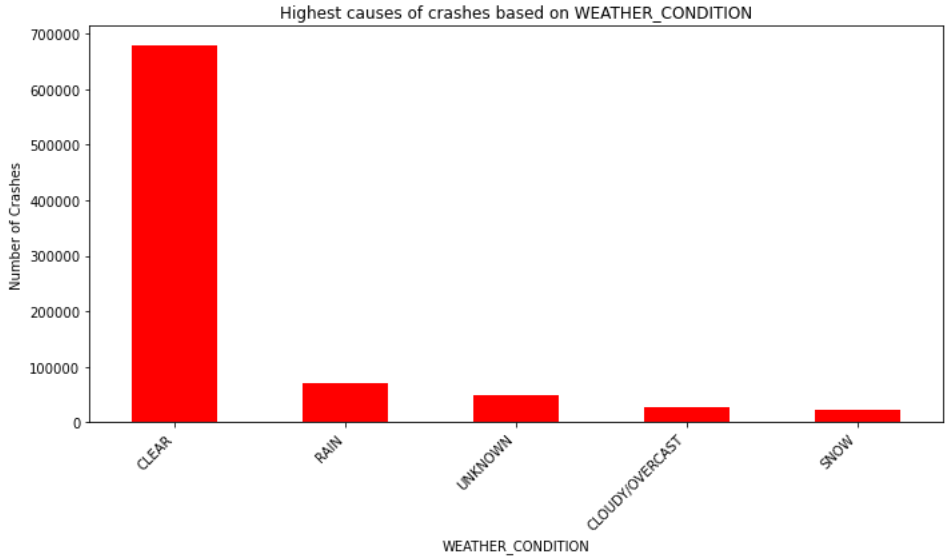
5.0 Exploratory Data Analysis

The primary contributory ca

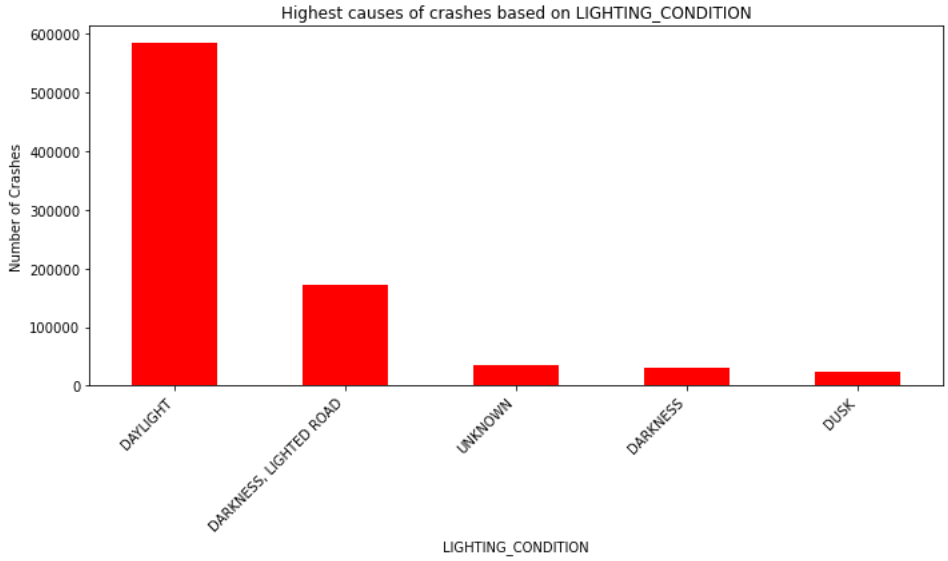
Findings

Most accidents:

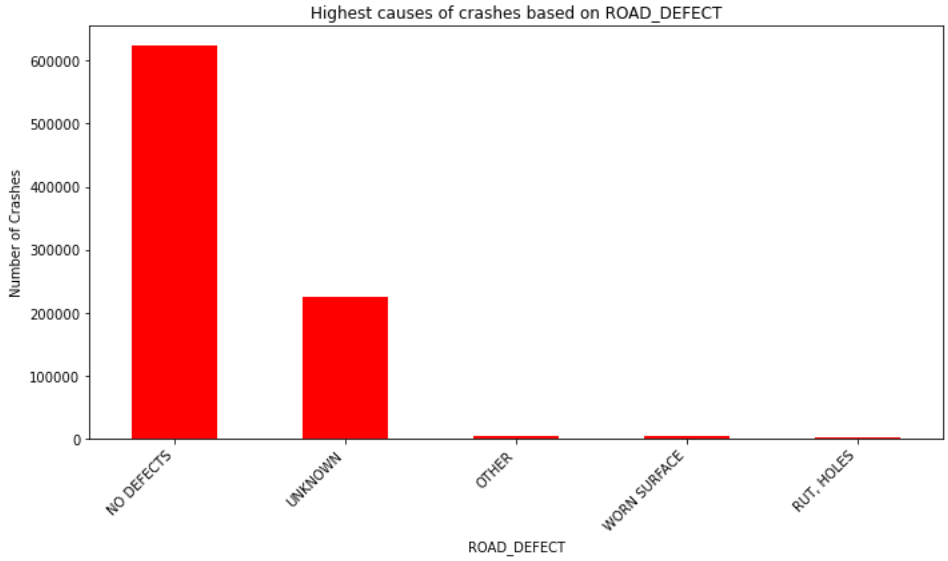
1. Occured when the weather was clear. However, when not clear, they occured in rainy conditions.



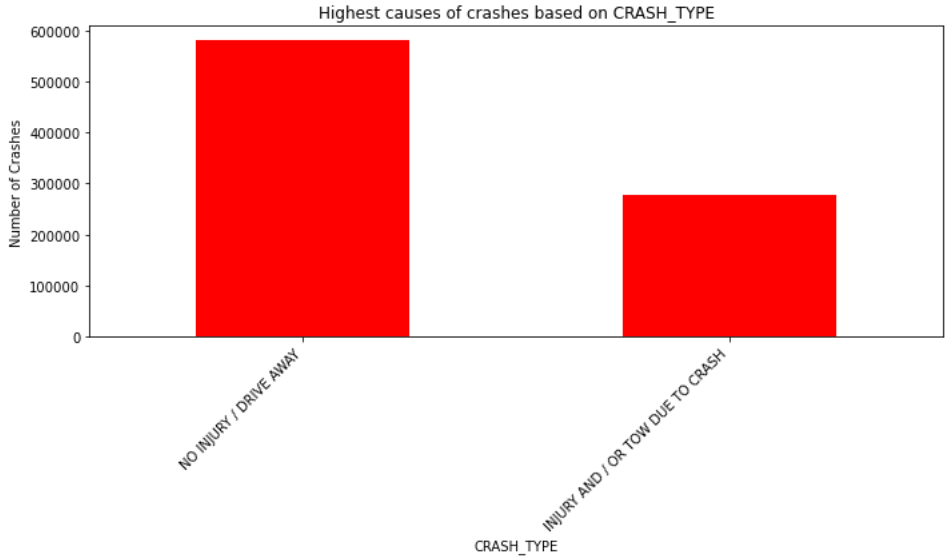
2. Occured in the daylight followed by darkly lighted roads.



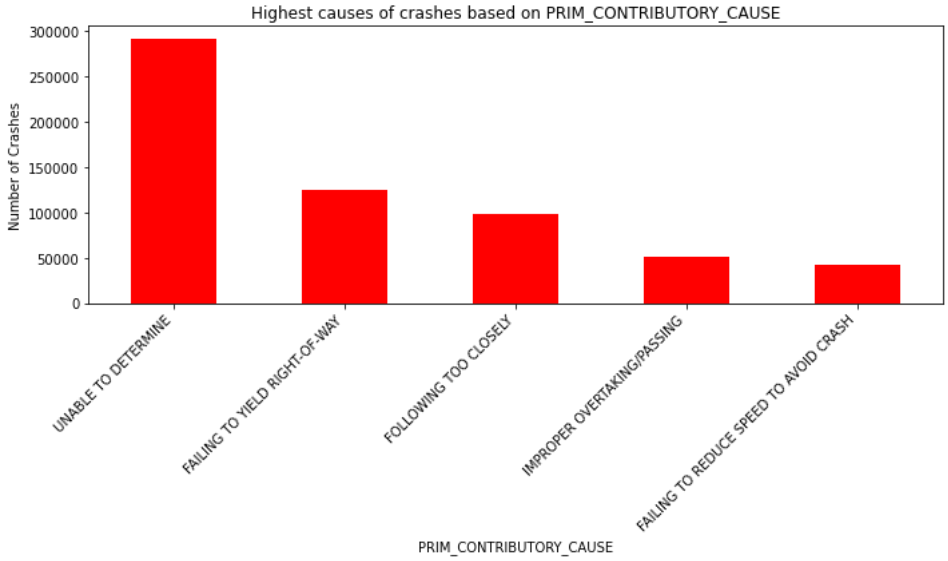
3. Occured where roads had no defects followed by roads with worn out surfaces and potholes.



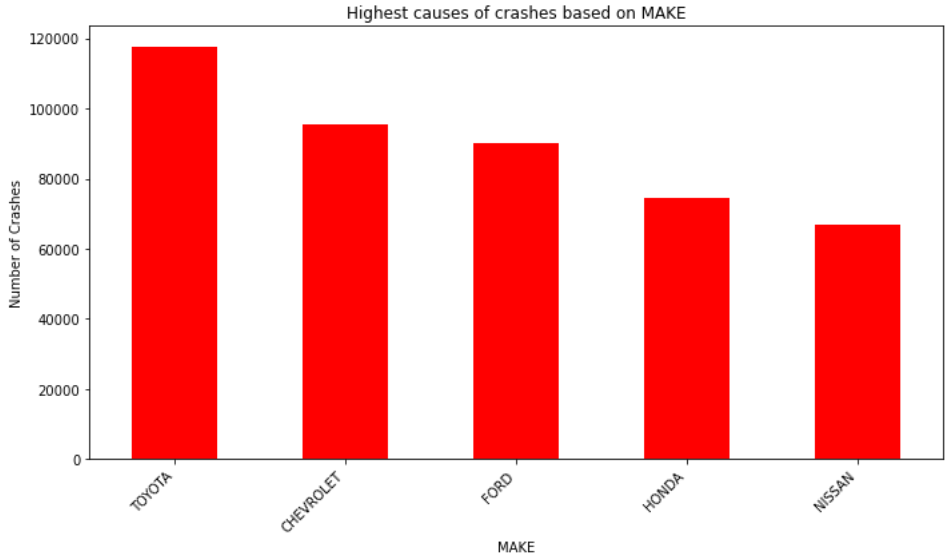
4. Did not involve any crash type injuries.

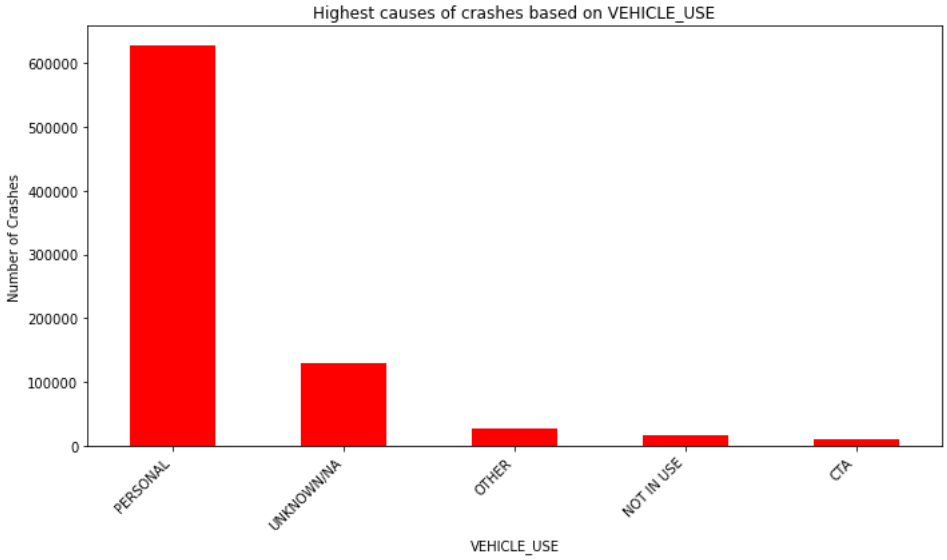


5. Did not have a primary cause but in those with a cause, the most common was either failing to give right of way or following too closely.

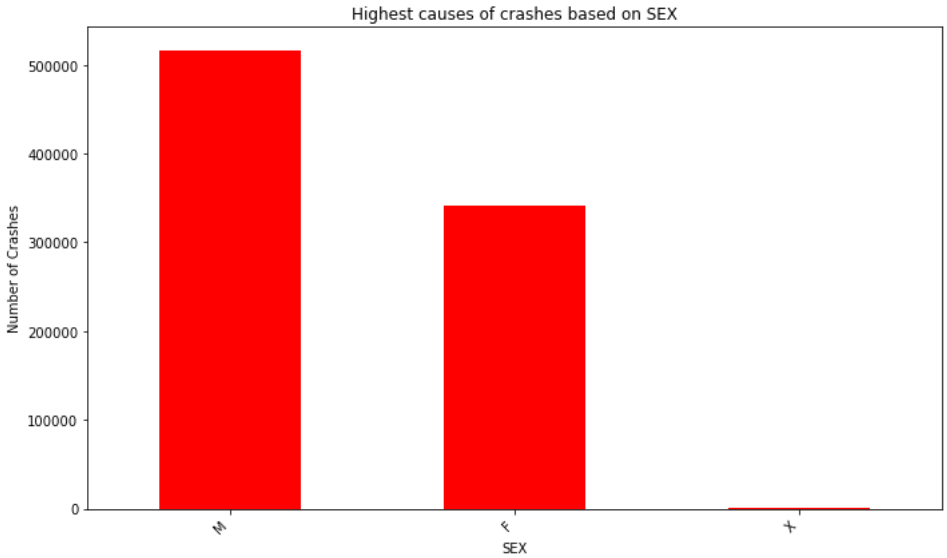


6. Involved Chevrolets, Fords and Nissans.



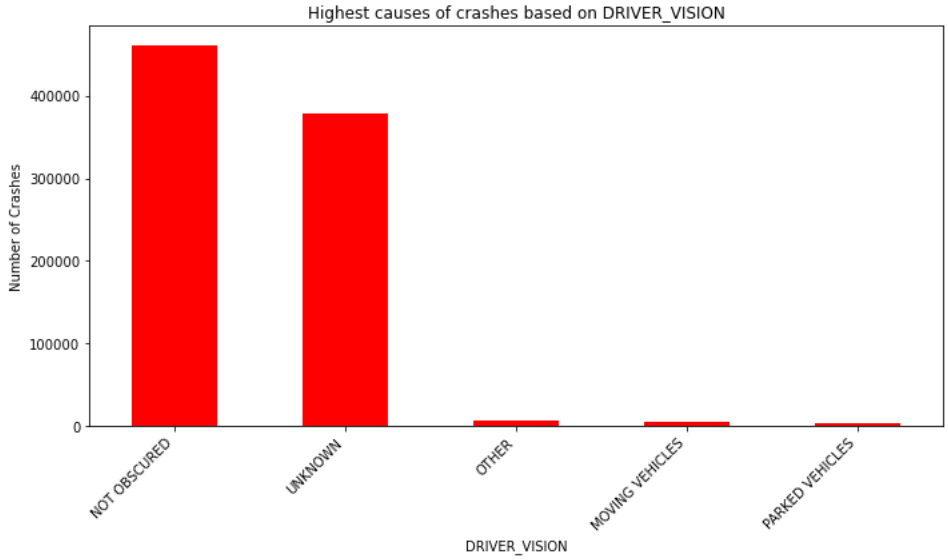


7. Involved personal cars.
8. Involved cars with no defects but of those which had brakes and tires had a defect.

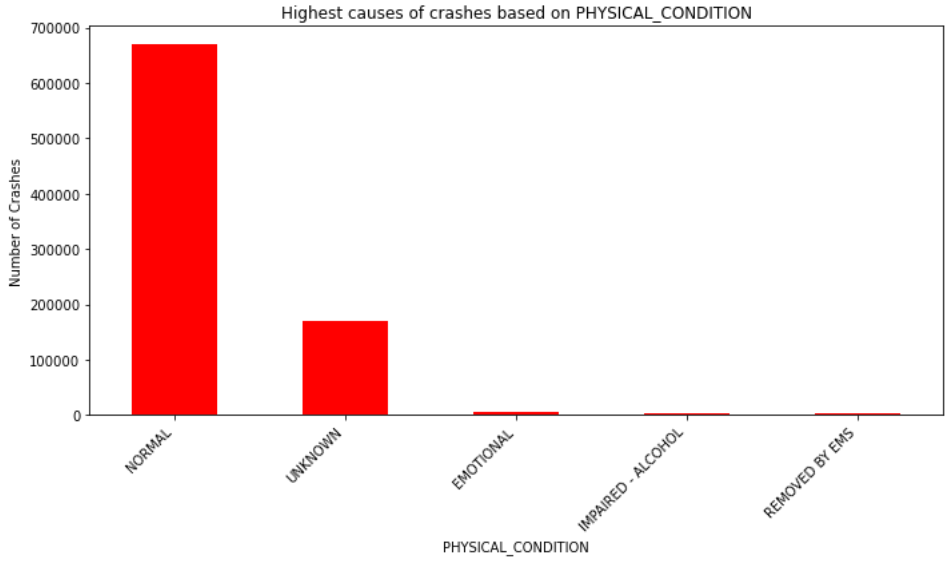


9. Involved more men than women.

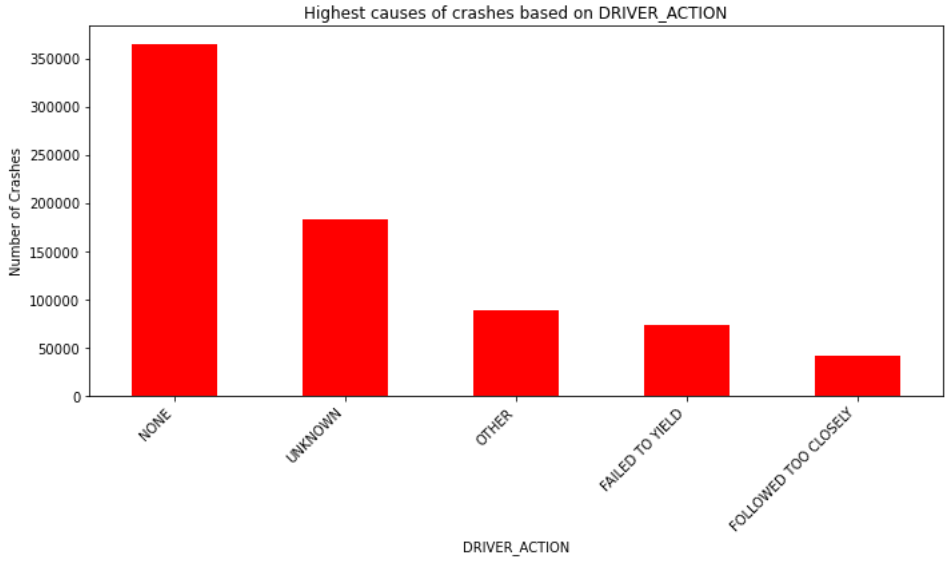
10. Were not caused by distractions, but of those were, the most common distractions were moving or parked vehicles.

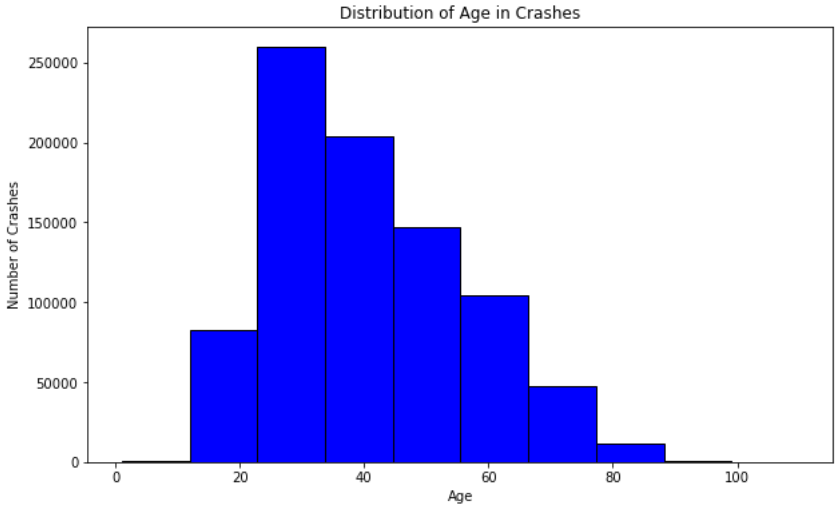


11. Involved drivers with normal physical conditions, or drivers with impaired alcohol levels, emotionally unstable or fatigued.



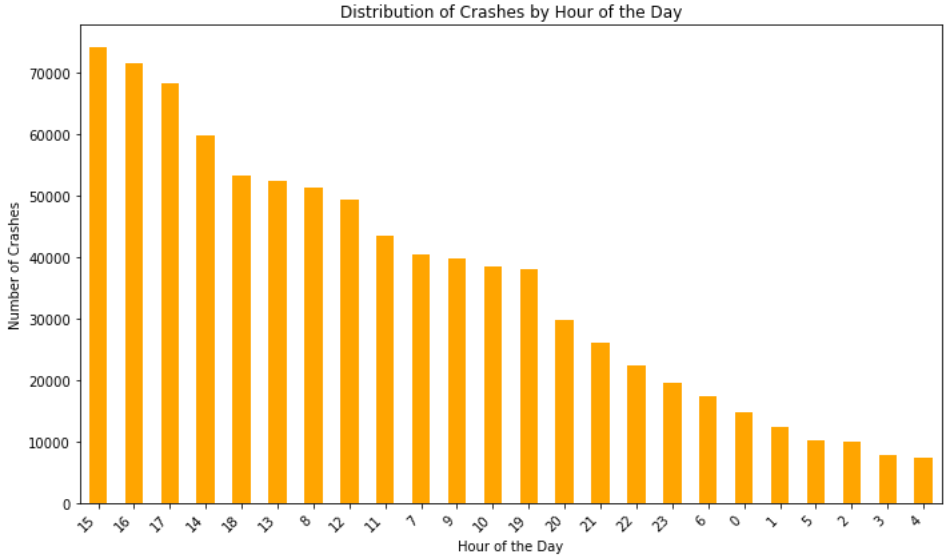
12. Had no preceding driver action. Of those which had, drivers either failed to yield or followed too closely.



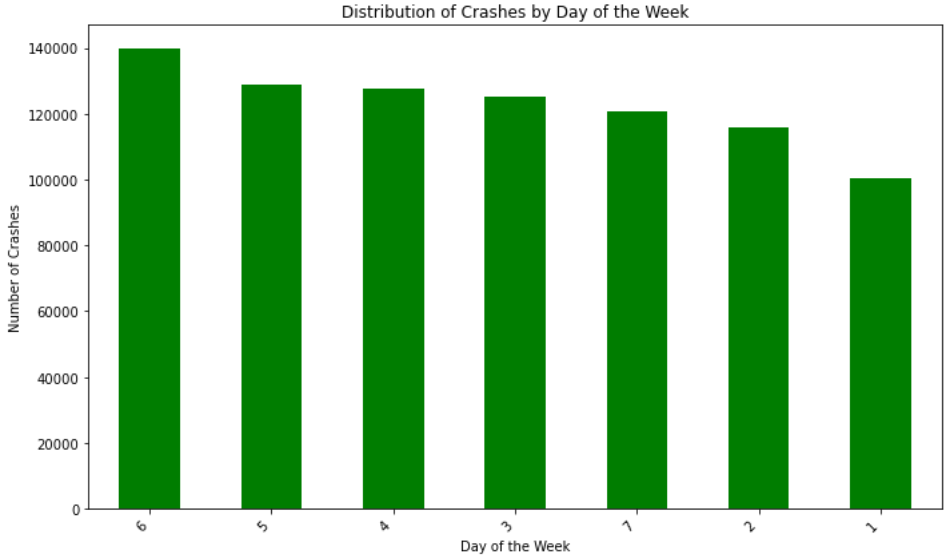


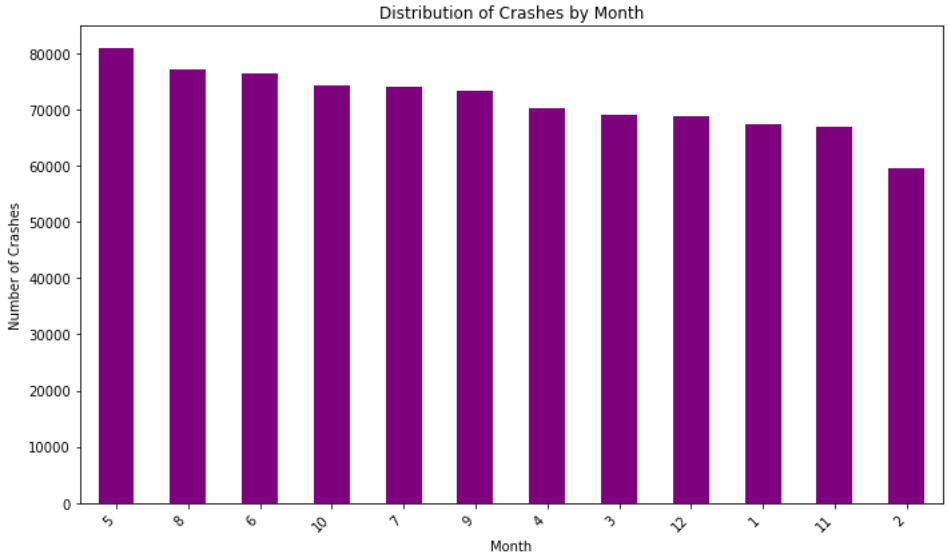
13. Involved drivers aged 20-40 years.

14. Occured between 3:00 p.m -5:00 p.m.



15. Occured on Fridays and Saturdays.





16. Occured in May and August.

Model Results

- We started by building baseline models including Decision Tree, Random Forest Classifier, and Neural Network

1. Initial models suffered from class imbalance by overfitting.

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 5

Languages

- Jupyter Notebook 100.0%